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Design for Repairability Demo

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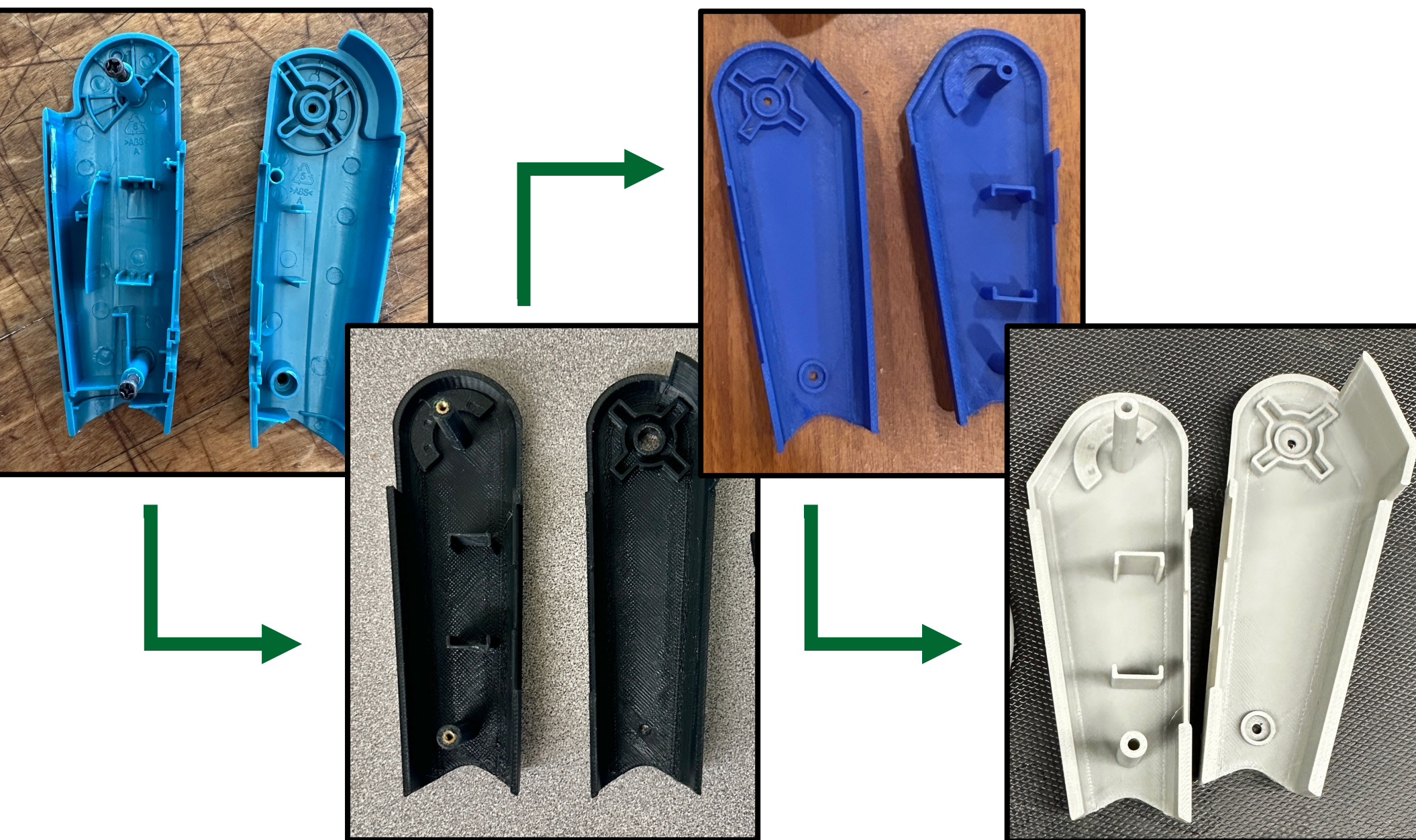


The Problem

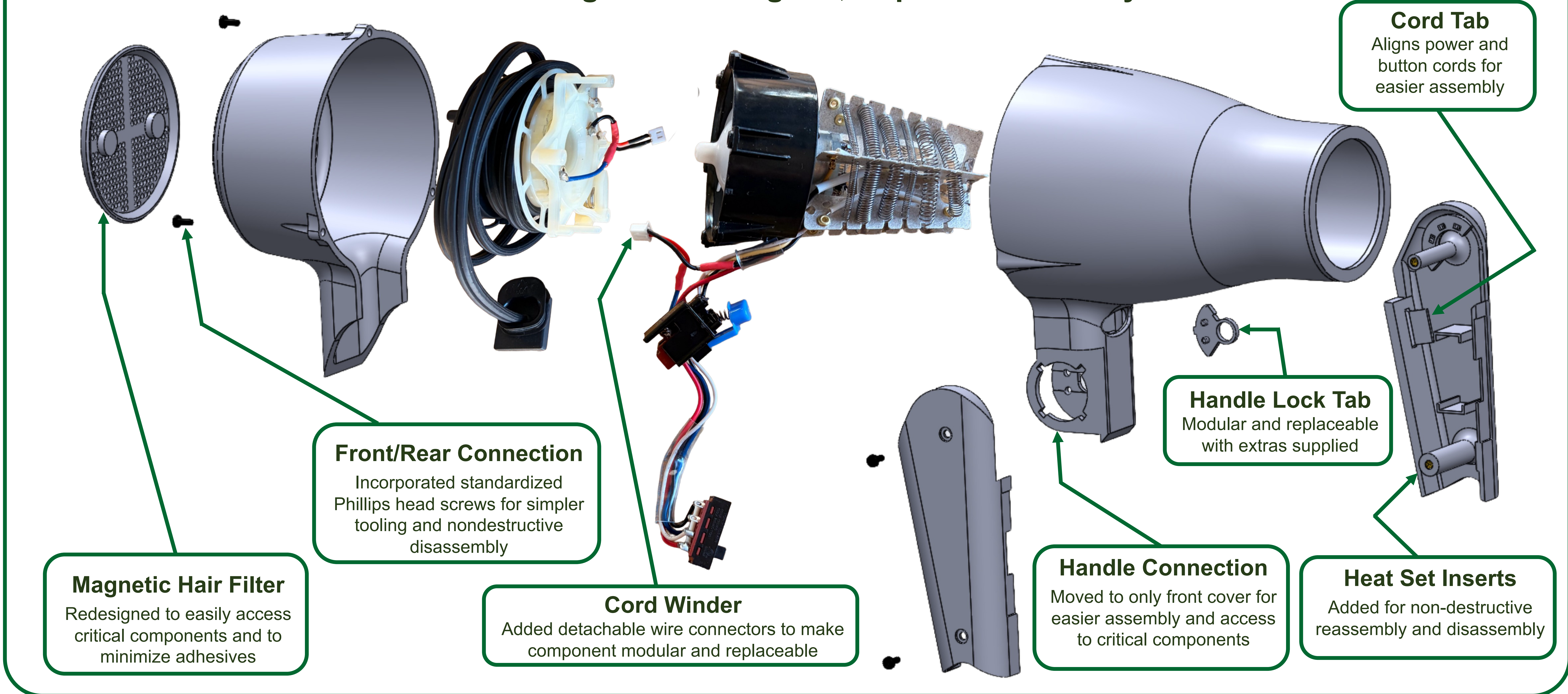
Design for repairability is a growing focus in modern product development due to widespread production of cheap electronic appliances with short service lives. The current ME curriculum does not directly address this facet of design, so Dr. Hyeonik Song is pioneering a new course within the Mechanical Engineering department. The Cal Poly Design for Repairability team was tasked with creating a suitable demonstration tool for use in the classroom, to help demonstrate the principles of design for repairability on a commercial appliance.

The Design Process

A key component of this design project was selecting an everyday device with poor repairability. We experimented with air fryers, lamps, pressure cookers, coffee makers, and many more appliances before settling on the Conair Cord-Keeper 1875W Compact hair dryer, pictured left. This unit features a retractable cord and a folding handle, two mechanisms which were found to be common failure modes from online reviews. The original model is difficult to disassemble, threaded holes are deformation prone, and accessibility of critical components is very poor, making it the perfect candidate to be redesigned. The design process required multiple iterations to optimize component fit and maintain original functionality (cord winding and folding handle). Several prior handle versions are shown below.



The Design: A Redesigned, Repairable Hair Dryer



Manufacturing

1. Print externals from PETG filament, pausing to insert magnets.

2. Apply 5 x M2.5 heat set inserts with the use of a soldering iron.

3. Solder on JST-XH wire connectors.

4. Cut power cord and press on the cap, applying superglue if necessary.

5. Insert existing internals and assemble device according to provided repair manual.

6. Screw together handle and covers using Phillips head M2.5 x 8 mm screws.

Testing

We conducted an in-class survey of Cal Poly engineering students to gauge whether they thought our redesigned hair dryer was repairable based on the given principles. We also performed internal testing to determine repeatability and ease of disassembly.

Table 1. Test results by test number.

Test #	Repairability Principle	Test Result	Pass/Fail
1	Access to Critical Components	Mean Score: 8.52/10	PASS
2	Modularity	Mean Score: 8.96/10	PASS
3	Adhesive Avoidance	Mean Score: 9.72/10	PASS
4	Easily Available Tools	% of Students w/ Phillips Head: 88%	PASS
5	Non-Destructive Disassembly	# of Broken Components: 0	PASS
6	Ease of Disassembly	Mean Disassembly + Assembly Time: 11.9 minutes	PASS

Next Steps

Moving forward, Dr. Song and other ME faculty will use this demonstration tool to illustrate key repairability concepts in ME 470: Designing Products for Repairability. This demo tool will withstand repeated use and, in the event of a breakage, is easily re-manufactured. Future course materials will likely include additional redesigned demos created by students enrolled in the class.

Acknowledgements

We would like to thank iFixit and the machine shop technicians at both Mustang '60 and the Mustang Makerspace for making this project possible.